

Software architecture

Software architecture

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Software architecture views

Documenting architecture views

Lecture: Documenting architecture views

Module: Software architecture views

What are architecture views?

- Representations of the overall architecture in terms meaningful to stakeholders.
- Enable the architecture to be communicated to and understood by the stakeholders.

Documenting architecture views

- The views of a software architecture are documented in many forms.
- Some elements are documented with text:
 - ✓ Risk mitigation plans,
 - ✓ Non-functional requirement in an SRS, etc.
- Other elements can be recorded using UML diagrams:
 - ✓ Component diagrams
 - ✓ Package diagrams
 - ✓ Deployment diagram

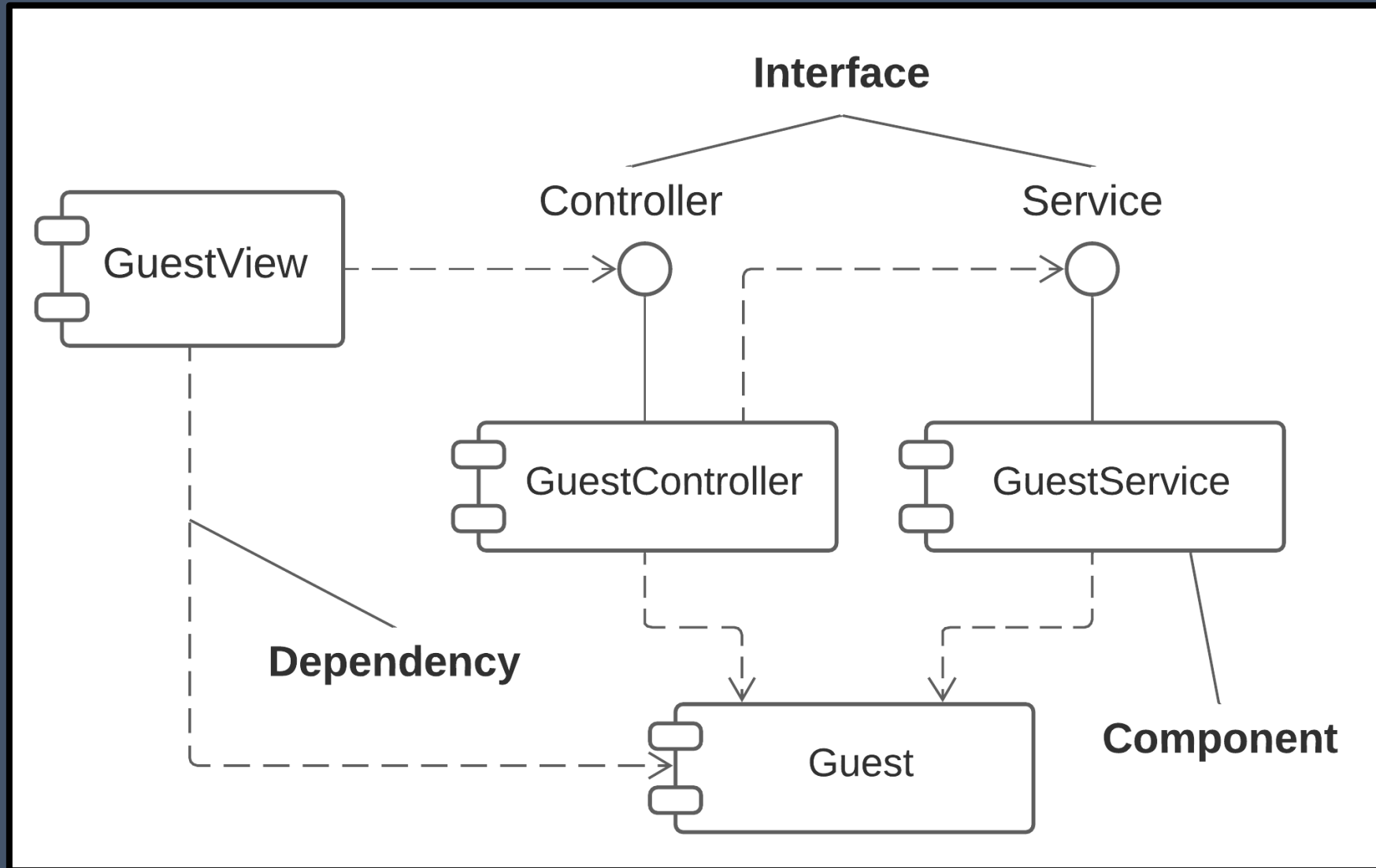
We focus on UML diagrams commonly used to document software architecture views

UML Component diagram

Lecture: UML Component diagram
Module: Software architecture views

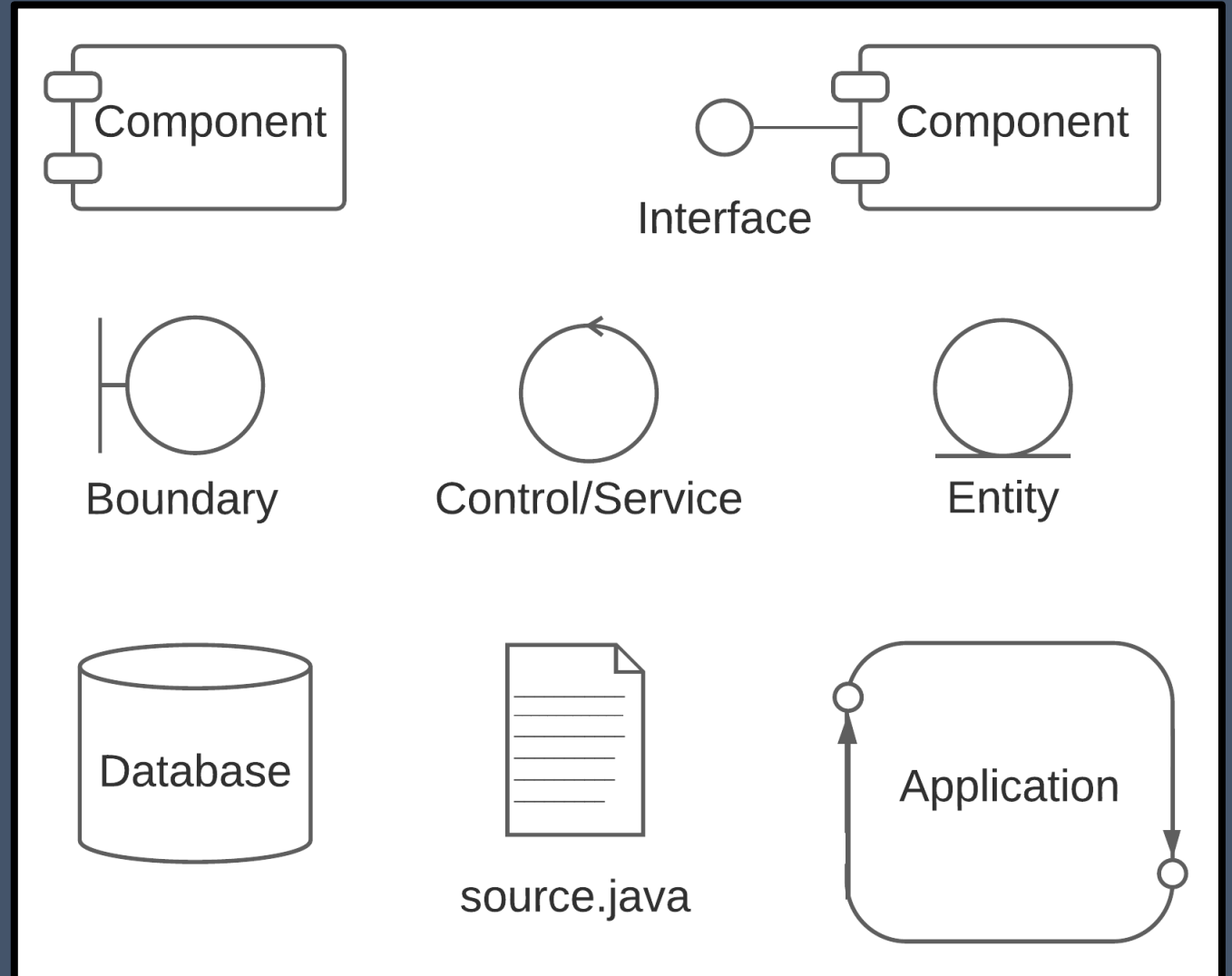
UML Component diagram

Elements of a UML Component diagram



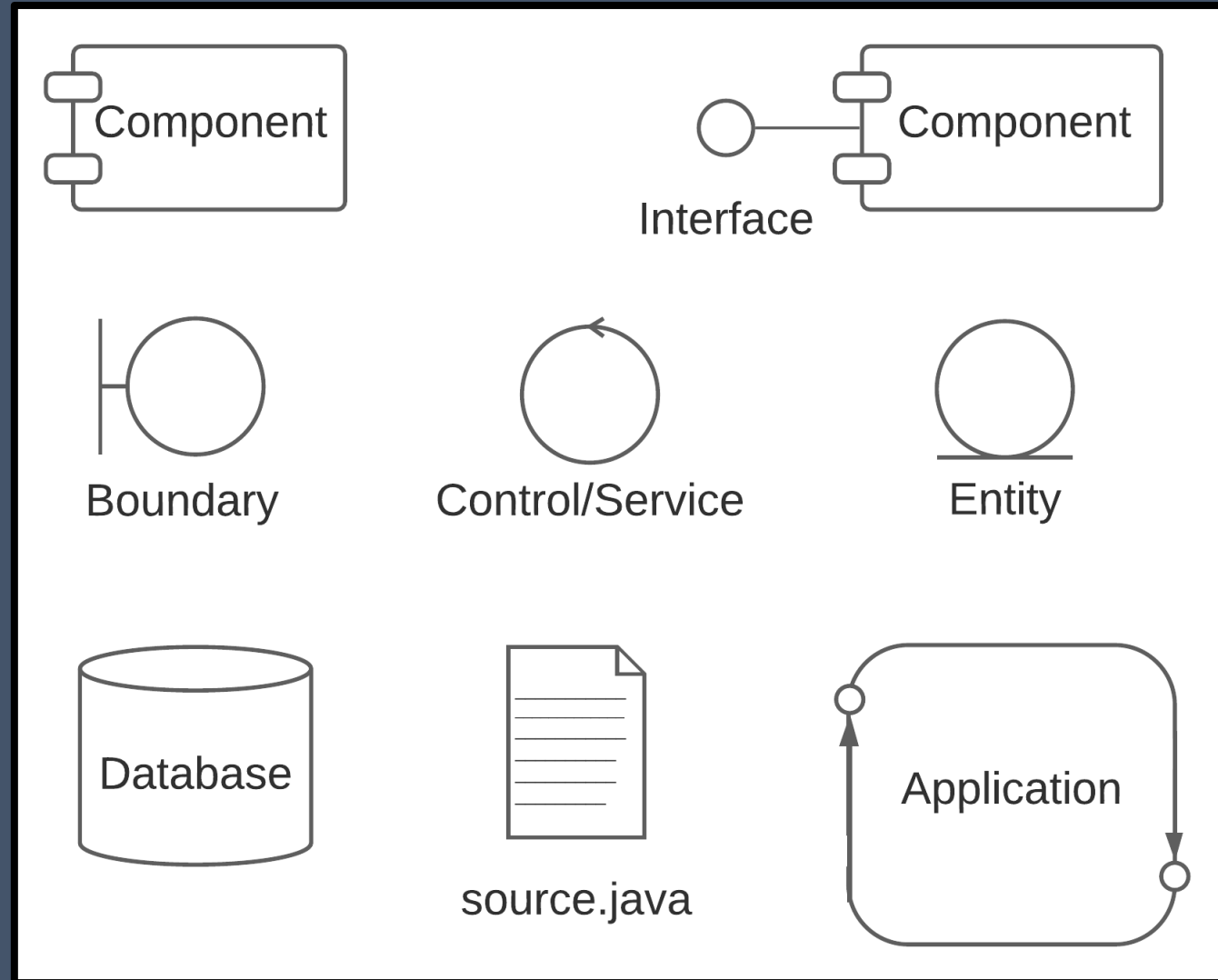
Characteristics and types of components

- A **component** is any **software unit**.
- Can be **large** and **abstract** or be **small** and **concrete**.



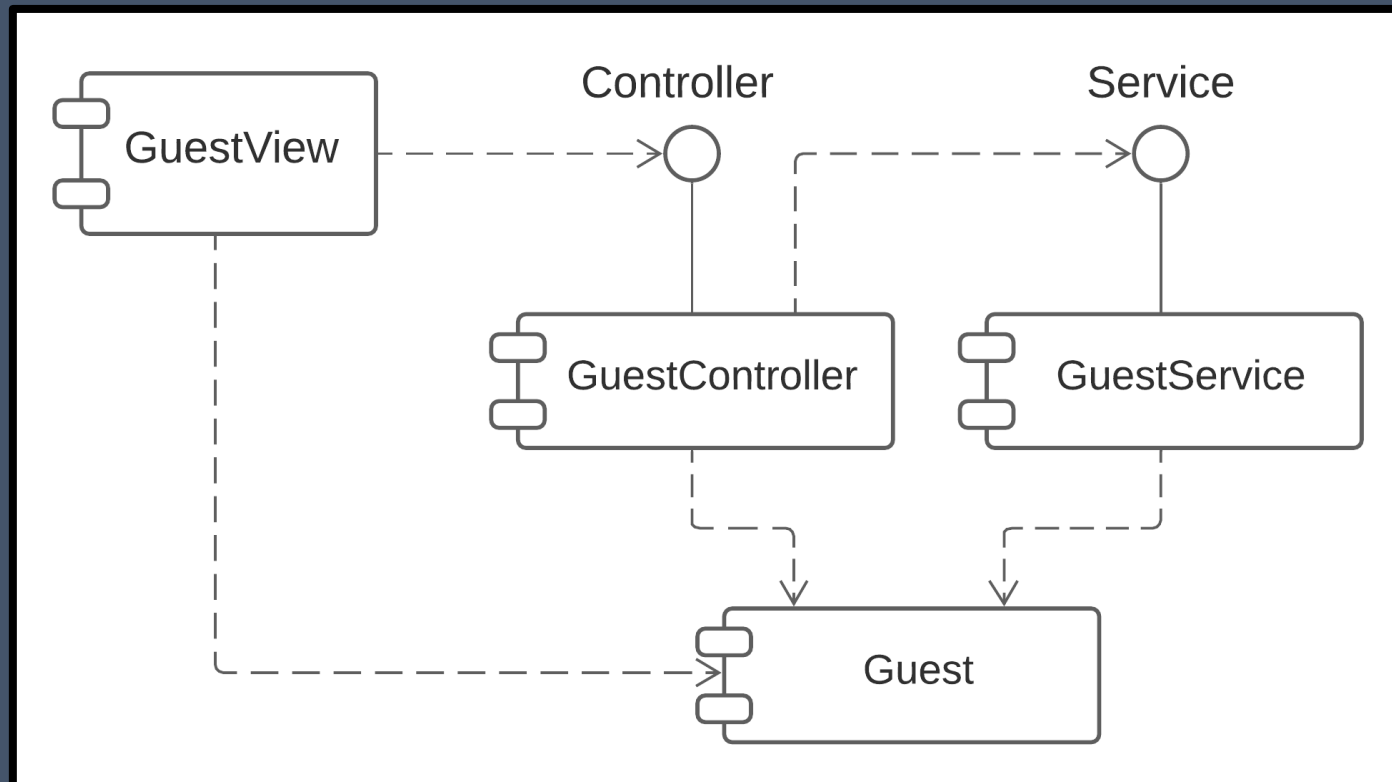
Characteristics and types of components

- May have an **interface** that it **exports** as a **service** to other **components**.
- Can be a **file**
 - ✓ **Source code** file, an **object file**, a complete **executable**, a **data file**, **HTML** or **media** files, and **so on**.



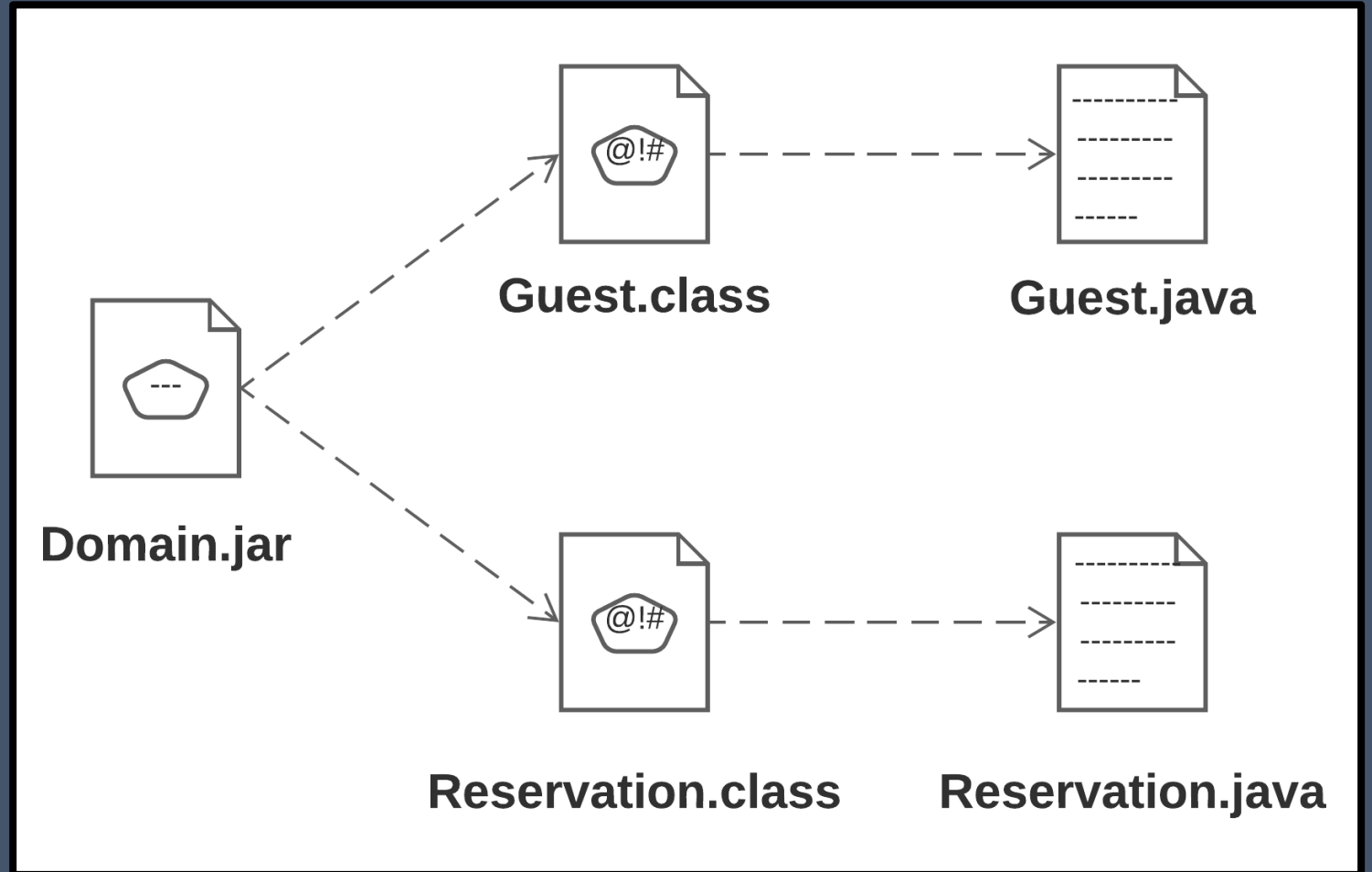
Example of a UML Component Diagram

- UML Component diagrams can show software dependencies.



Build structures

- **Component diagrams** can represent build structures.



UML Package diagram

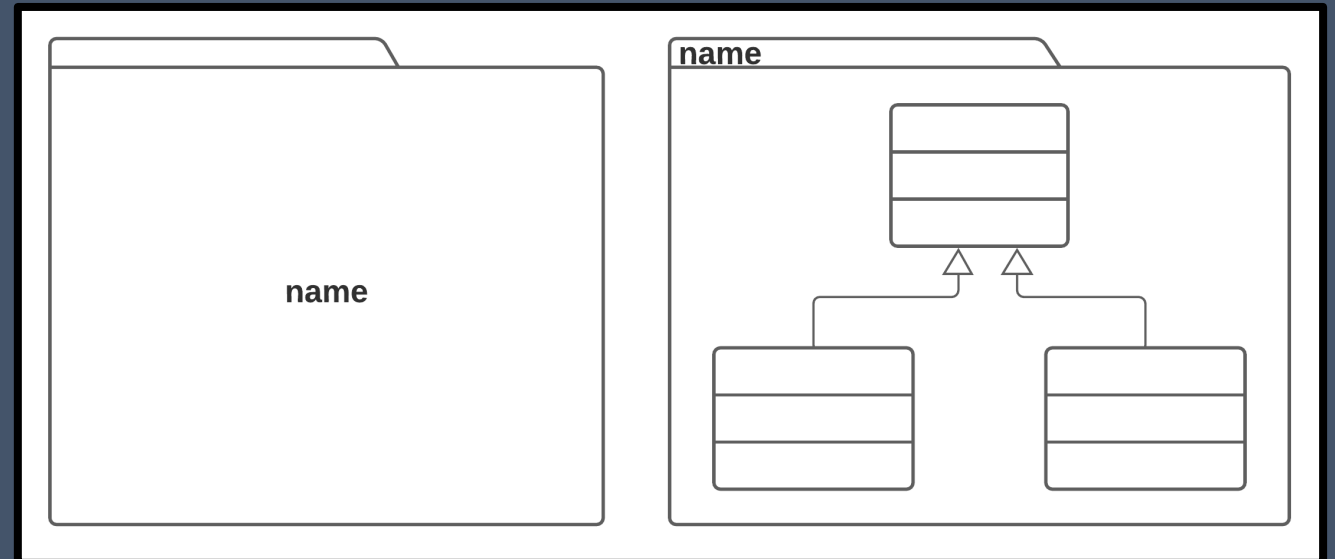
Lecture: UML Package diagram

Module: Software architecture views

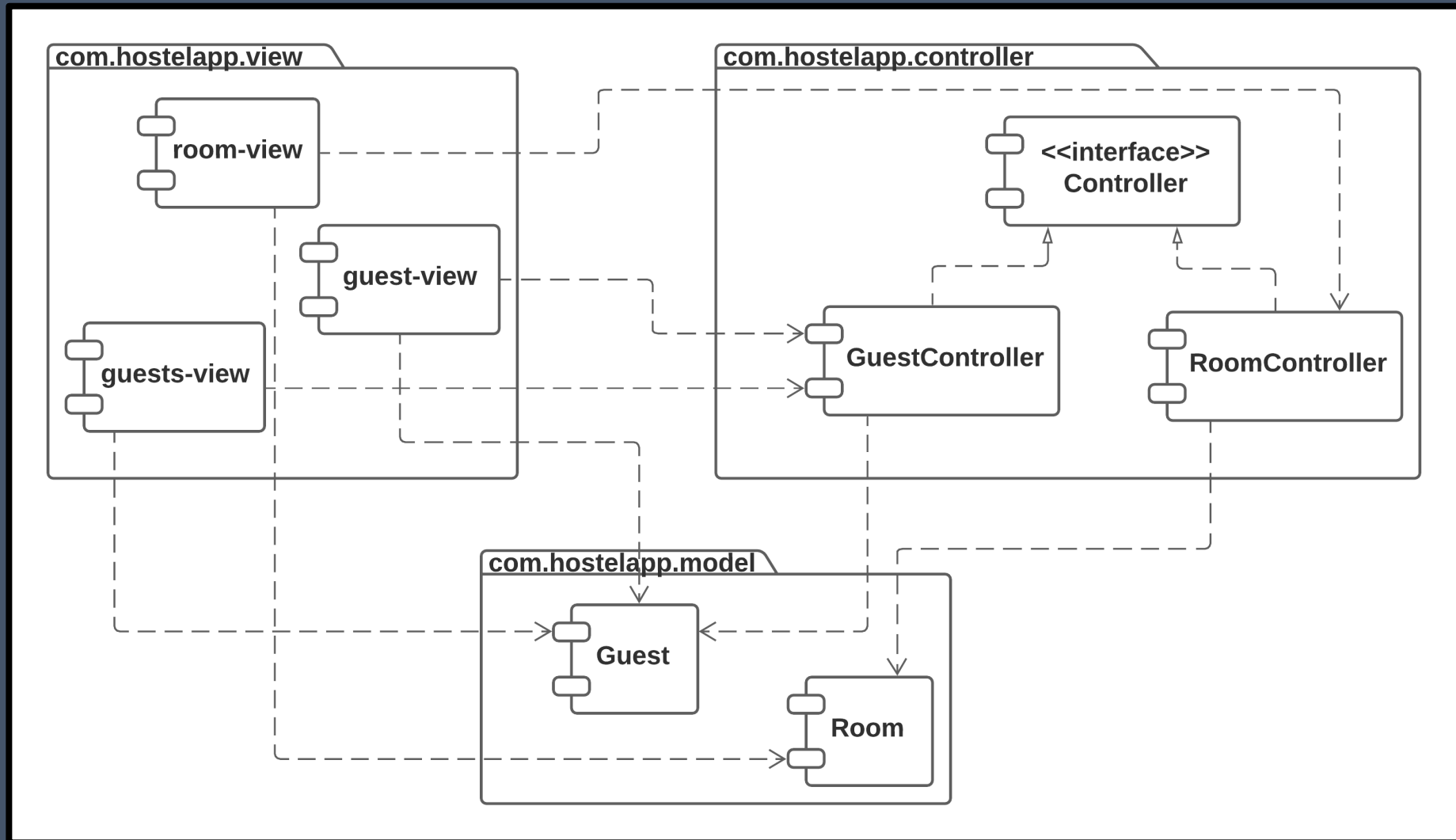
UML Package diagram

Elements of a UML Package diagram

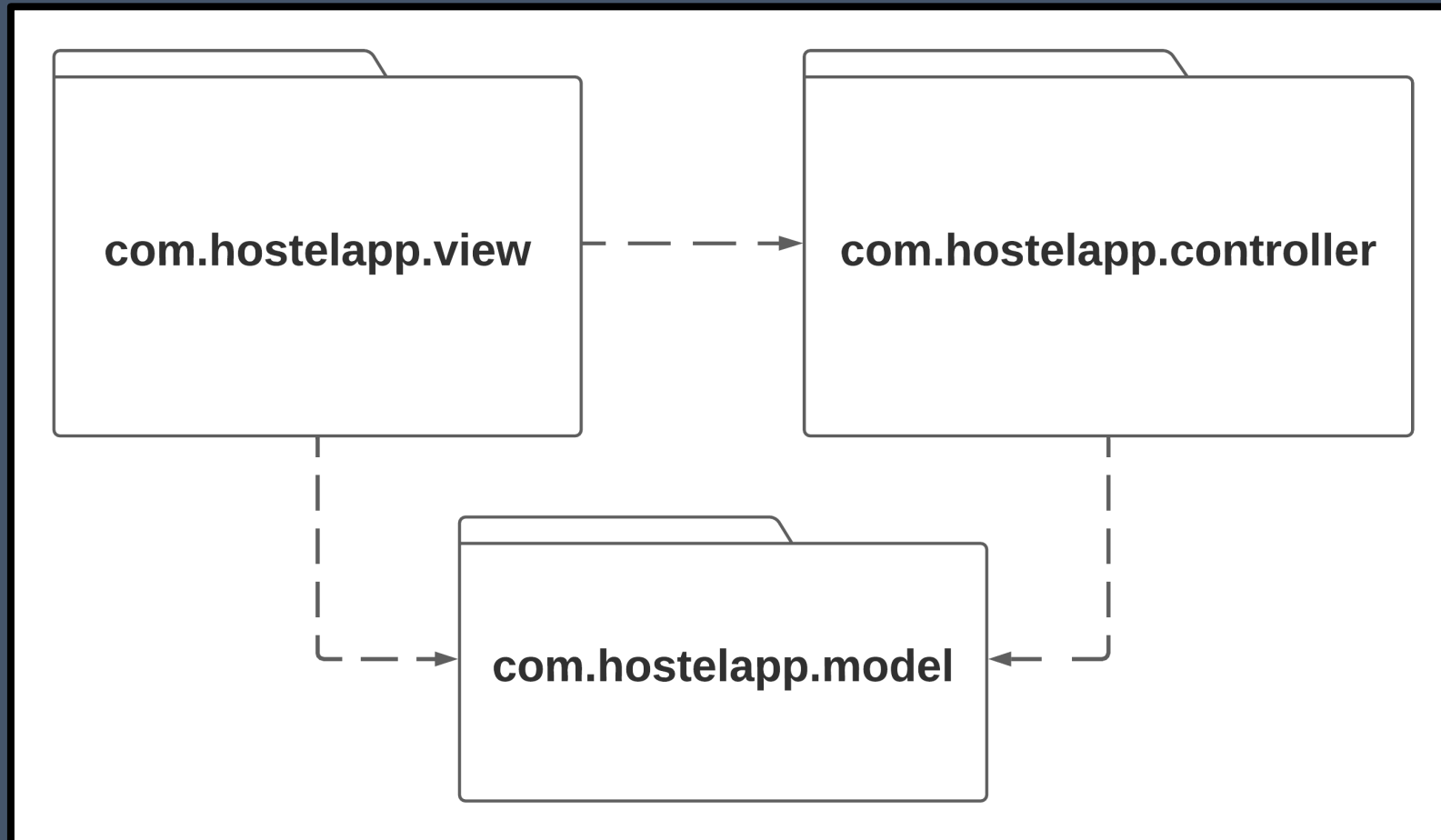
- A UML Package diagram shows dependencies between packages.
 - ✓ A package can hold any UML element.
- The package name can be placed in the body box or in the name box.



Example of a UML Package diagram



Abstract UML Package diagram



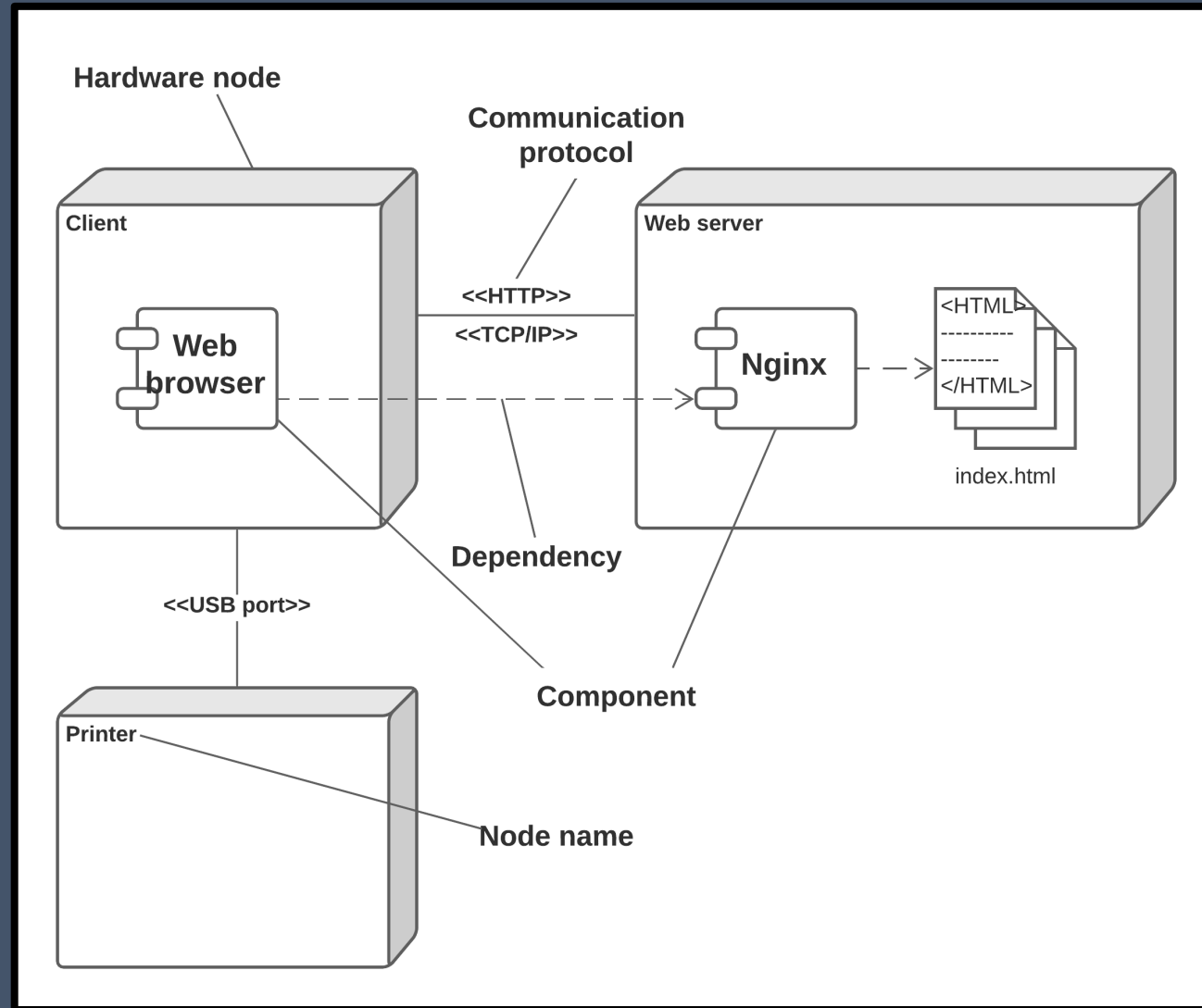
UML Deployment diagram

Lecture: UML Deployment diagram

Module: Software architecture views

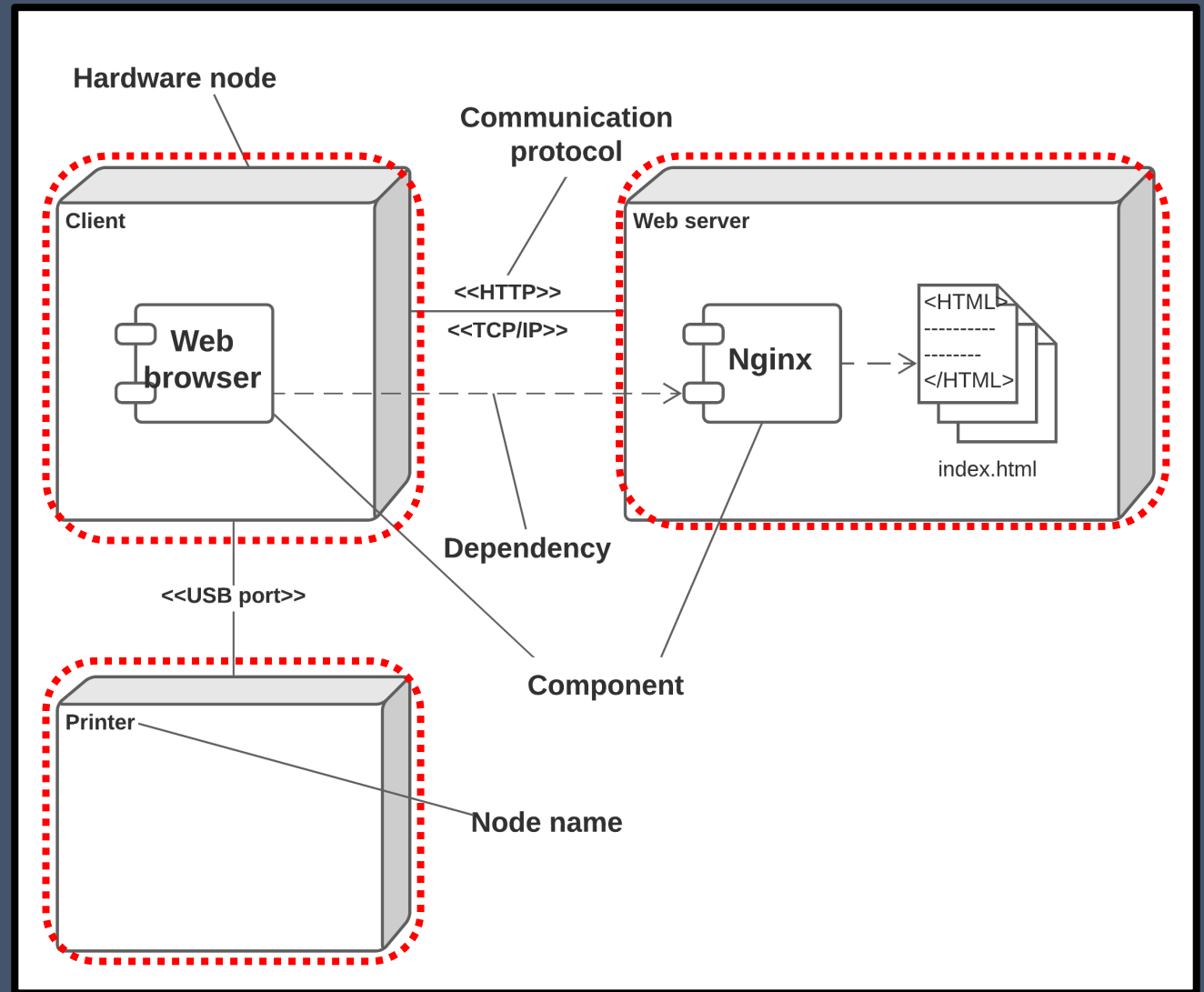
UML Deployment diagram

Elements of a UML Deployment Diagram



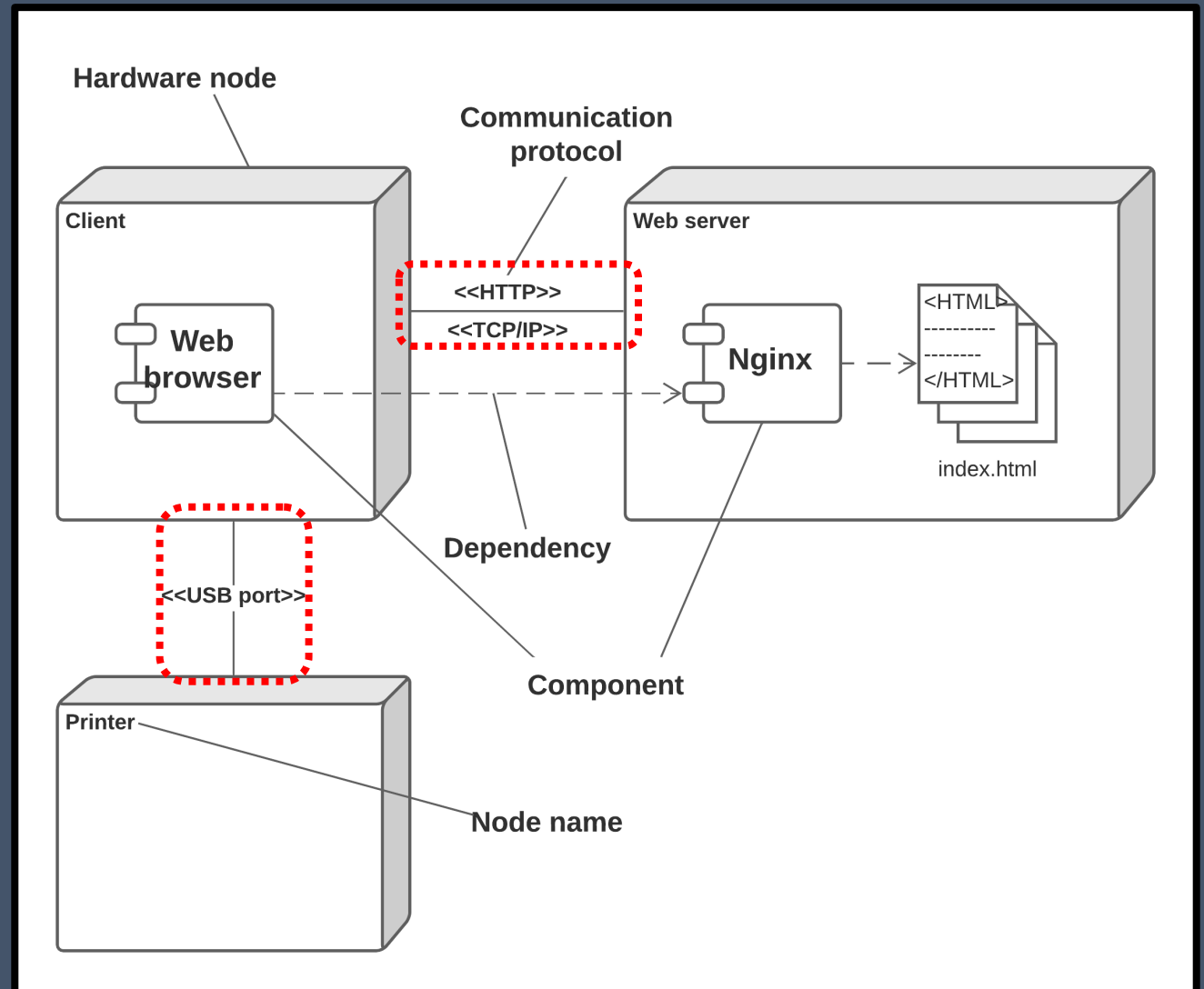
Elements of a UML Deployment Diagram

- Hardware nodes can represent any type of hardware.



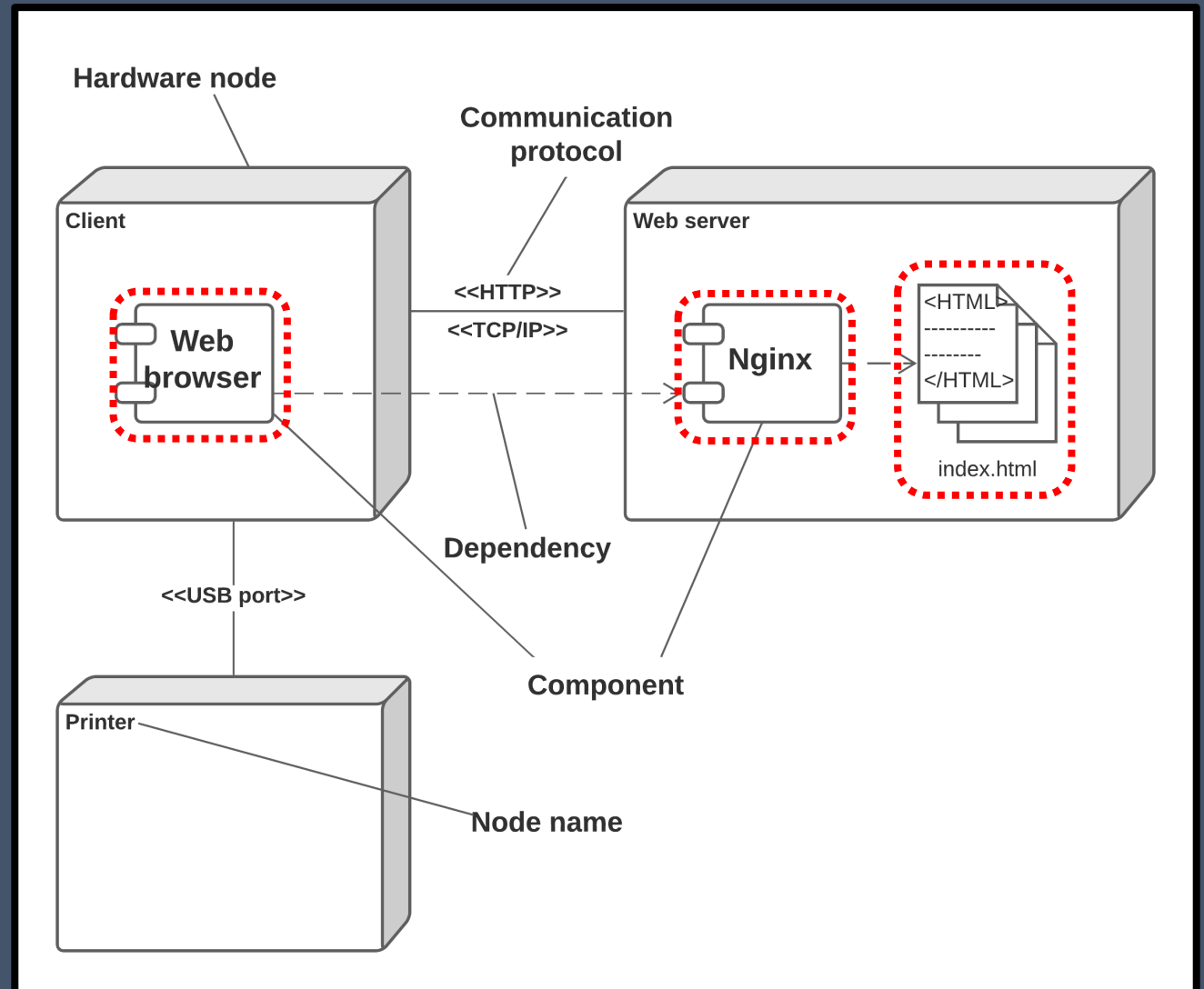
Elements of a UML Deployment Diagram

- Links between hardware nodes indicate connectivity
 - ✓ Can also include the communication protocol used between nodes.

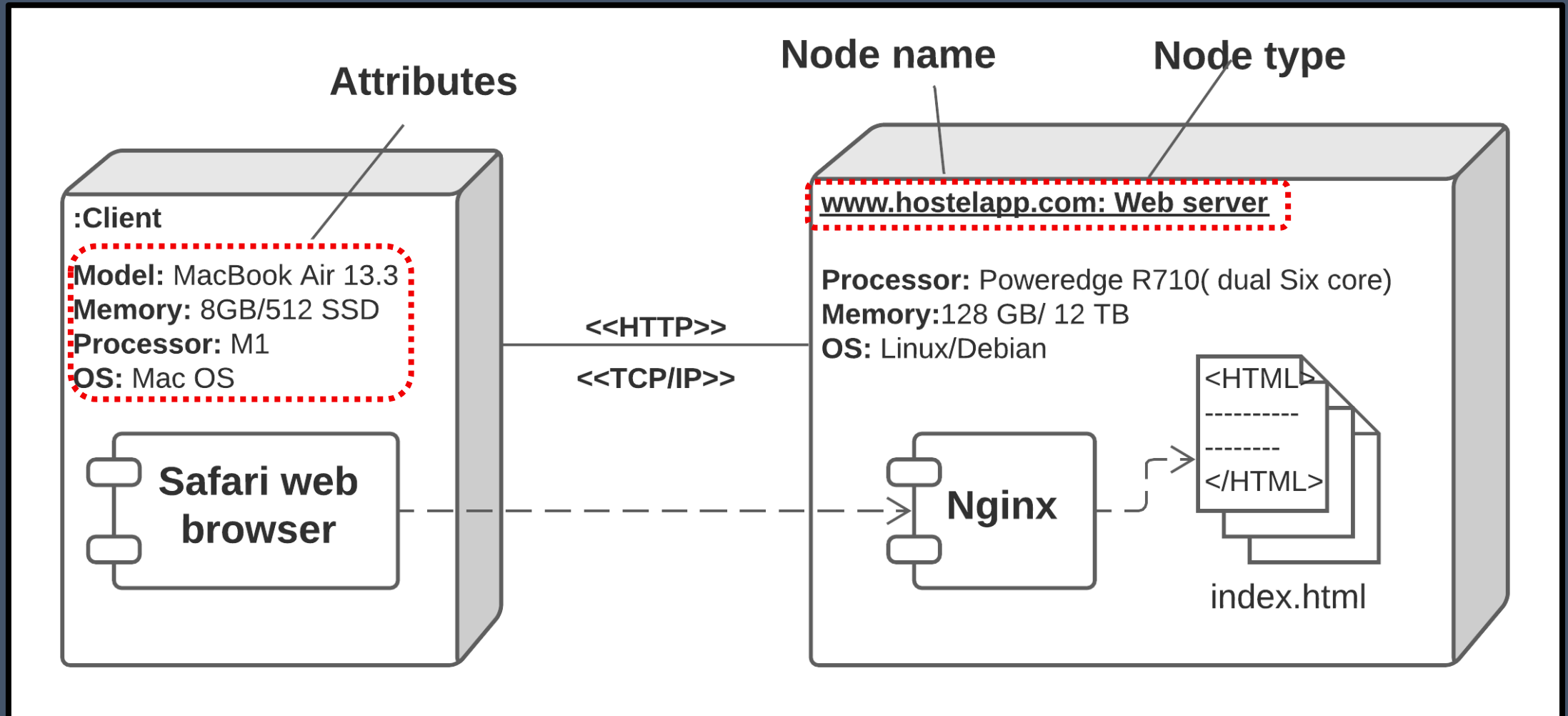


Elements of a UML Deployment Diagram

- Software components are placed within hardware nodes to show the distribution of the software across the network.



Additional elements: attributes & node type



Questions?